

Reading Scaled Picture and Bar Graphs

Answer Key

Grade 3

Quick Answers

- | | |
|---------------|-------------------------|
| 1. 15 | 5. 30 |
| 2. 30 | 6. 8 |
| 3. (a) = 5, — | 7. birds |
| 4. 12 | 8. (a) = 4, 6, 8, 12, — |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.MD – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 10 tagged checks.

- 1.** Each book symbol stands for 5 books. Ana’s row shows 3 symbols. How many books did Ana read?

Solution: Each symbol counts 5 books, so multiply the symbols by the scale: $3 \times 5 = 15$.

Ana read

15

books.

- 2.** Ben’s row shows 4 symbols and Cara’s row shows 2 symbols. How many books did they read together?

Solution: Together they show $4 + 2 = 6$ symbols, and each symbol is 5 books: $6 \times 5 = 30$.

Ben and Cara read

30

books together.

- 3.** Dev read 25 books. (a) How many symbols show 25 books? (b) Draw Dev’s row.

Solution (a): Each symbol is 5 books, so divide: $25 \div 5 = 5$. Dev’s row needs

5

symbols.

Model answer (b), instructor judges: The drawn row must show exactly 5 book symbols — the same number found in part (a). Any clear book-symbol drawing counts.

- 4.** Use Graph B (the number line counts by twos). How many dogs are there?

Solution: The dogs bar reaches the gridline labeled 12 — six steps of 2 above zero. There are 12 dogs.

5. How many pets does the whole graph show in all?

Solution: Read each bar against the scale, then add every group: $8 + 12 + 6 + 4 = 30$. The graph shows 30 pets in all.

6. How many more dogs are there than birds?

Solution: “How many more” compares the two bars, so subtract: $12 - 4 = 8$. There are 8 more dogs than birds.

7. Which pet group is the smallest?

Solution: The shortest bar is the smallest group. The bars read 8, 12, 6, and 4, and the shortest is the birds bar at 4. The smallest group is birds

8. (a) Write the four pet counts in order from least to greatest. (b) Which display makes comparing easier — picture graph or bar graph? Say why.

Solution (a): The four counts are 8, 12, 6, and 4. From least to greatest: 4, 6, 8, 12

Model answer (b), instructor judges: Either display can earn credit with a fitting reason — for example: the bar graph, because taller and shorter bars can be compared at a glance without counting; or the picture graph, because the symbols can be counted one by one. The answer must name a display and give a why.